- Data Transfer and Manipulation
  - Most computer instructions can be classified into three categories:
    - 1) Data transfer, 2) Data manipulation, 3) Program control instructions
  - Data Transfer Instruction
    - Typical Data Transfer Instruction : ***Tab. 8-5***
      - **Load** : transfer from memory to a processor register, usually an AC (*memory read*)
      - **Store** : transfer from a processor register into memory (*memory write*)
      - **Move** : transfer from one register to another register
      - **Exchange** : swap information between two registers or a register and a memory word
      - **Input/Output** : transfer data among processor registers and input/output device
      - **Push/Pop** : transfer data between processor registers and a memory stack
    - 8 Addressing Mode for the LOAD Instruction : Tab. 8-6
      - @ : Indirect Address
      - **$** : Address relative to PC
      - # : Immediate Mode
      - **( )** : Index Mode, Register Indirect, Autoincrement register
  - Data Manipulation Instruction
    - 1) Arithmetic, 2) Logical and bit manipulation, 3) Shift Instruction

TABLE 8-5 Typical Data Transfer Instructions

Name	Mnemonic
Load	LD
Store	ST
Move	MOV
Exchange	XCH
Input	IN
Output	OUT
Push	PUSH
Pop	POP

TABLE 8-6 Eight Addressing Modes for the Load Instruction

Mode	Assembly Convention	Register Transfer
Direct address	LD ADR	$AC \leftarrow M[ADR]$
Indirect address	LD @ADR	$AC \leftarrow M[M[ADR]]$
Relative address	LD $ADR	$AC \leftarrow M[PC + ADR]$
Immediate operand	LD #NBR	$AC \leftarrow NBR$
Index addressing	LD ADR(X)	$AC \leftarrow M[ADR + XR]$
Register	LD R1	$AC \leftarrow R1$
Register indirect	LD (R1)	$AC \leftarrow M[R1]$
Autoincrement	LD (R1)+	$AC \leftarrow M[R1],\ R1 \leftarrow R1 + 1$

TABLE 8-10 Typical Program Control Instructions

Name	Mnemonic
Branch	BR
Jump	JMP
Skip	SKP
Call	CALL
Return	RET
Compare (by subtraction)	CMP
Test (by ANDing)	TST

- Arithmetic Instructions : ***Tab. 8-7***
- Logical and Bit Manipulation Instructions : ***Tab. 8-8***
- Shift Instructions : ***Tab. 8-9***

- **Program Control**
  - Program Control Instruction : ***Tab. 8-10***
    - Branch and Jump instructions are used interchangeably to mean the same thing
  - Status Bit Conditions : ***Fig. 8-8***
    - Condition Code Bit or Flag Bit
      - The bits are set or cleared as a result of an operation performed in the ALU
  - 4-bit status register
    - Bit **C** (*carry*) : set to 1 if the end carry C_8 is 1
    - Bit **S** (*sign*) : set to 1 if F_7 is 1
    - Bit **Z** (*zero*) : set to 1 if the output of the ALU contains all 0's
    - Bit **V** (*overflow*) : set to 1 if the exclusive-OR of the last two carries (C_8 and C_7) is equal to 1
    - ***Flag Example*** : A - B = A + (2's Comp. Of B) : ***A =11110000, B = 00010100***

11110000
+ *11101100* (2's comp. of B)
1 11011100

C = 1, S = 1, V = 0, Z = 0

TABLE 8-7 Typical Arithmetic Instructions

Name	Mnemonic
Increment	INC
Decrement	DEC
Add	ADD
Subtract	SUB
Multiply	MUL
Divide	DIV
Add with carry	ADDC
Subtract with borrow	SUBB
Negate (2's complement)	NEG

TABLE 8-8 Typical Logical and Bit Manipulation Instructions

Name	Mnemonic
Clear	CLR
Complement	COM
AND	AND
OR	OR
Exclusive-OR	XOR
Clear carry	CLRC
Set carry	SETC
Complement carry	COMC
Enable interrupt	EI
Disable interrupt	DI

TABLE 8-9 Typical Shift Instructions

Name	Mnemonic
Logical shift right	SHR
Logical shift left	SHL
Arithmetic shift right	SHRA
Arithmetic shift left	SHLA
Rotate right	ROR
Rotate left	ROL
Rotate right through carry	RORC
Rotate left through carry	ROLC

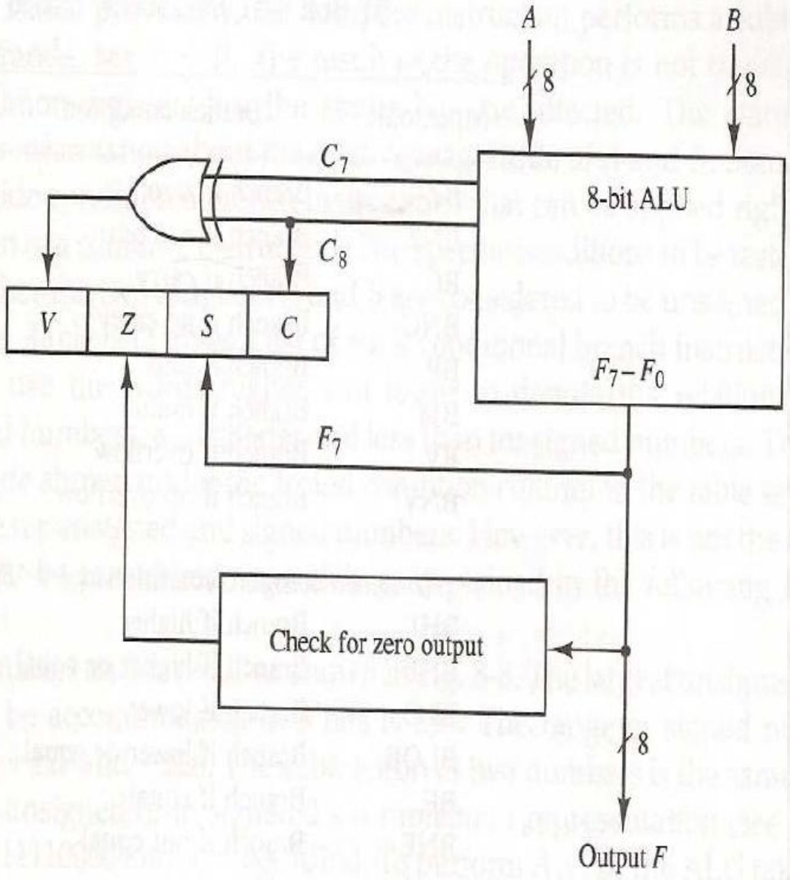

Figure 8-8 Status register bits.

TABLE 8-11 Conditional Branch Instructions

Mnemonic	Branch condition	Tested condition
BZ	Branch if zero	$Z = 1$
BNZ	Branch if not zero	$Z = 0$
BC	Branch if carry	$C = 1$
BNC	Branch if no carry	$C = 0$
BP	Branch if plus	$S = 0$
BM	Branch if minus	$S = 1$
BV	Branch if overflow	$V = 1$
BNV	Branch if no overflow	$V = 0$
	Unsigned compare conditions $(A - B)$	
BHI	Branch if higher	$A > B$
BHE	Branch if higher or equal	$A \geq B$
BLO	Branch if lower	$A < B$
BLOE	Branch if lower or equal	$A \leq B$
BE	Branch if equal	$A = B$
BNE	Branch if not equal	$A \neq B$
	Signed compare conditions $(A - B)$	
BGT	Branch if greater than	$A > B$
BGE	Branch if greater or equal	$A \geq B$
BLT	Branch if less than	$A < B$
BLE	Branch if less or equal	$A \leq B$
BE	Branch if equal	$A = B$
BNE	Branch if not equal	$A \neq B$